Lin Tian

Postdoctoral Research Fellow, Massachusetts General Hospital & Harvard Medical School
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EDUCATION

University of North Carolina at Chapel Hill, North Carolina, U.S. 2019 — 2024
Ph.D. in Computer Science
Supervisor: Prof. Marc Niethammer
Thesis: Learning Generalizable Deformations From Images

University of Southern California, California, U.S. 2010 — 2012
M.S. in Computer Science

Huazhong University of Science and Technology, Hubei, China 2006 — 2010
B.Eng. in Software Engineering

RESEARCH EXPERIENCE

Massachusetts General Hospital / Harvard Medical School Boston, U.S.
Research Fellow, Supervisor: Dr. Juan Eugenio Iglesias and Dr. Matthew Rosen September 2024 — Present

Department of Computer Science, University of North Carolina at Chapel Hill Chapel Hill, U.S.
Research Assistant, Supervisor: Dr. Marc Niethammer August 2019 — July 2024

Google X, Alphabet Inc. Mountain View, U.S.
PhD Residency, Supervisor: Dr. Alexander Zoellner, Dr. Ningrui Li and Dr. Atilla Kiraly May 2023 — August 2023

Damo Academy, Alibaba Group New York, U.S.
Research Scientist Intern, Supervisor: Dr. Dakai Jin, Dr. Ke Yan and Dr. Ling Zhang May 2022 — August 2022

AI Lab, ByteDance Ltd. Mountain View, U.S.
Research Scientist Intern, Supervisor: Dr. Imran Saleemi May 2021 — August 2021

Ruijia Technology Inc. Wuhan, China
Machine Learning Engineer, Supervisor: Dr. Rong Yuan March 2017 — August 2019

PUBLICATIONS

* Equal Contribution

Selected Preprints & Papers Under Review

[1] **Lin Tian**, Xiaoling Hu, and Juan Eugenio Iglesias. "Test-time Uncertainty Estimation for Medical Image Registration via Transformation Equivariance". Under Review at **Top-Tier AI Conference**. arXiv:2509.23355. 2025.

[2] **Lin Tian**, Jonathan Williams Ramirez, Dina Zemlyanker, Lucas J. Deden-Binder, Rogeny Herisse, Theresa R. Connors, Mark Montine, Istvan N. Huszar, Lilla Zöllei, Sean I. Young, Christine Mac Donald, C. Dirk Keene, Derek H. Oakley, Bradley T. Hyman, Oula Puonti, Matthew S. Rosen, and Juan Eugenio Iglesias. "Reference-Free 3D Reconstruction of Brain Dissection Slabs via Learned Atlas Coordinates". arXiv:2503.09963. 2025.

[3] **Lin Tian***, Zi Li*, Fengze Liu, Xiaoyu Bai, Jia Ge, Le Lu, Marc Niethammer, Xianghua Ye, Ke Yan, and Daikai Jin. "SAME++: A Self-supervised Anatomical eMbeddings Enhanced Medical Image Registration Framework using Stable Sampling and Regularized Transformation". arXiv:2311.14986. 2024.

Peer-reviewed Conference Papers

[1] Hastings Greer, **Lin Tian**, Francois-Xavier Vialard, Roland Kwitt, Raul San Jose Estepar, and Marc Niethammer. "CARL: A Framework for Equivariant Image Registration". In: **CVPR** (2025).

[2] Başar Demir, **Lin Tian**, Hastings Greer, Roland Kwitt, François-Xavier Vialard, Raúl San José Estépar, Sylvain Bouix, Richard Rushmore, Ebrahim Ebrahim, and Marc Niethammer. "Multigradicon: A Foundation Model for Multimodal Medical Image Registration". In: International Workshop on Biomedical Image Registration (WBIR) at MICCAI (2024). (**Best Oral Presentation**).

[3] **Lin Tian**, Hastings Greer, Roland Kwitt, Francois-Xavier Vialard, Raul San Jose Estepar, Sylvain Bouix, Richard Rushmore, and Marc Niethammer. "uniGradICON: A Foundation Model for Medical Image Registration". In: **MICCAI** (2024). (**Over 8,600 Downloads**).

[4] **Lin Tian**, Roni Sengupta, Hastings Greer, Raúl San José Estépar, and Marc Niethammer. "NePhi: Neural Deformation Fields for Approximately Diffeomorphic Medical Image Registration". In: **ECCV** (2024).

[5] Hastings Greer, **Lin Tian**, Francois-Xavier Vialard, Roland Kwitt, Sylvain Bouix, Raul San Jose Estepar, Richard Rushmore, and Marc Niethammer. "Inverse Consistency by Construction for Multistep Deep Registration". In: **MICCAI** (2023).

[6] Zi Li*, **Lin Tian***, Tony CW Mok, Xiaoyu Bai, Puyang Wang, Jia Ge, Jingren Zhou, Le Lu, Xianghua Ye, Ke Yan, and Dakai Jin. "Samconvex: Fast Discrete Optimization for CT Registration using Self-supervised Anatomical Embedding and Correlation Pyramid". In: **MICCAI** (2023).

[7] **Lin Tian***, Hastings Greer*, François-Xavier Vialard, Roland Kwitt, Raúl San José Estépar, Richard Jarrett Rushmore, Nikolaos Makris, Sylvain Bouix, and Marc Niethammer. "GradICON: Approximate Diffeomorphisms via Gradient Inverse Consistency". In: **CVPR** (2023).

[8] **Lin Tian**, Yueh Z Lee, Raúl San José Estépar, and Marc Niethammer. "LiftReg: Limited Angle 2D/3D Deformable Registration". In: **MICCAI** (2022).

[9] Peirong Liu, **Lin Tian**, Yubo Zhang, Stephen Aylward, Yueh Lee, and Marc Niethammer. "Discovering Hidden Physics behind Transport Dynamics". In: **CVPR** (2021).

[10] **Lin Tian**, Connor Puett, Peirong Liu, Zhengyang Shen, Stephen R Aylward, Yueh Z Lee, and Marc Niethammer. "Fluid Registration between Lung CT and Stationary Chest Tomosynthesis Images". In: **MICCAI** (2020).

[11] **Lin Tian** and Rong Yuan. "An Automatic End-to-end Pipeline for CT Image-based EGFR Mutation Status Classification". In: Medical imaging 2019: image processing. Vol. 10949. SPIE. 2019, pp. 695–700.

[12] Rong Yuan, **Lin Tian**, and Junhui Chen. "An RF-BFE Algorithm for Feature Selection in Radiomics Analysis". In: Medical Imaging 2019: Imaging Informatics for Healthcare, Research, and Applications. Vol. 10954. SPIE. 2019, pp. 183–188.

Technical Reports & Community Benchmarks

[1] Bhakti Baheti, Satrajit Chakrabarty, Hamed Akbari, ..., **Lin Tian**, Hastings Greer, Marc Niethammer, ..., Spyridon Bakas, and Diana Waldmannstetter. "The Brain Tumor Sequence Registration (BraTS-Reg) Challenge: Establishing Correspondence Between Pre-Operative and Follow-up MRI Scans of Diffuse Glioma Patients". arXiv:2112.06979. 2024.

PATENTS

Zi Li, **Lin Tian**, Tony C. W. Mok, Xiaoyu Bai, Puyang Wang, Le Lu, Ke Yan, Dakai Jin. "Image Registration Method, Electronic Device, and Computer-Readable Storage Medium". Chinese Patent CN116823905B, Granted.

OPEN SOURCE SOFTWARE

uniGradICON (Medical Foundation Model) `[GitHub][PyPI]` 2024

A general-purpose medical image registration foundation model with zero-shot capabilities across anatomies, modalities, and deformation patterns. Released as a Python package (**over 22,000 PyPI downloads**) and a **3D Slicer Extension** to enable clinical deployment.

HONORS & AWARDS

- **Rising Stars in EECS**, MIT & Boston University 2025
  Selected as one of the top upcoming researchers in EECS from over 300 applicants (Acceptance Rate: $\sim 21\%$).

- **NSF Travel Award**, CVPR Doctoral Consortium 2025
  Competitive selection (Acceptance Rate: ∼18% worldwide).
- **Best Oral Presentation**, MICCAI WBIR Workshop 2024
- **Student Travel Award**, MICCAI 2022
- **Apple Design Award** (Member of the engineering team) 2012
  Selected as one of the three awarded iPhone apps worldwide (for "Where's My Water?") by Apple.
- **University Merit Scholarship**, Huazhong University of Science and Technology 2008
  Awarded to the top 1% of students.
- **National Scholarship**, Ministry of Education 2007
  Top 0.2% of undergraduates nationwide in China.
- **University Merit Scholarship**, Huazhong University of Science and Technology 2007
  Awarded to the top 1% of students.

ACADEMIC SERVICES

Area Chair
MICCAI 2026
MICCAI 2025

Conference Reviewer
CVPR, ECCV, ICCV, MICCAI, AAAI, ICML, NeurIPS

Journal Reviewer
Medical Image Analysis
IEEE Transactions on Medical Imaging
IEEE Transactions on Pattern Analysis and Machine Intelligence
Pattern Recognition

TEACHING EXPERIENCES

University of Southern California Los Angeles, U.S.
Teaching Assistant | CSCI-526 Advanced Mobile Devices & Games Spring 2012

MENTORSHIP

Wenzhu Ye	2025—2026	M.Sc.	Harvard University	Co-mentored with Prof. Iglesias on super-resolution models for dental imaging
Zhaoxi Zhang	2023—2024	M.Sc.	UNC Chapel Hill	Reconstructing CT from Single-view and Biplanar X-Rays
Yueliang Ying	2023—2024	M.Sc.	UNC Chapel Hill	Reconstructing CT from Single-view and Biplanar X-Rays

INDUSTRY ENGINEERING & PRODUCT EXPERIENCE

Lilith Games Shanghai, China
Virtual Reality Game Producer & Senior System Designer May 2015 — Feb 2017

- **R&D Leadership (PongIt! VR):** Led a cross-functional team to develop a VR sports title on Steam, managing the full lifecycle from prototype to launch.
- **Data Science (Soul Hunters):** Architected data mining scripts to analyze user telemetry for the top-grossing game of 2014, optimizing retention strategies for millions of users.

Netease Games Hangzhou, China
System Designer June 2014 — May 2015

- **Gameplay Design:** Designed and prototyped core combat algorithms for mobile RPGs.

Disney Interactive Media Group Los Angeles, U.S.
Game Development Engineer June 2012 — May 2014

- **Rendering Optimization (Where's My Water?):** Optimized the rendering pipeline within the proprietary game engine, improving frame rates and visual quality for the **2012 Apple Design Award** winner.
- **Infrastructure & DevOps:** Engineered CI/CD pipelines using Jenkins and Git to automate build testing; developed internal tools to accelerate asset deployment for the art team.